Teaching Portfolio- Philipp Stark

1. Teaching Philosophy

My teaching philosophy is grounded in the belief that university teaching should empower students to think critically, develop independent learning strategies, and acquire skills that they can apply beyond the classroom. I see myself not primarily as an examiner, but as a tutor and guide: someone who creates a safe, stimulating environment where students are encouraged to ask questions, explore alternative perspectives, and connect theory with practice.

Central to this approach are three guiding principles:

1. **Student-centred learning**: Students learn most effectively when they are actively engaged. I use interactive activities, group projects, and problem-based exercises to create learning processes where students are co-producers of knowledge (Kugel, 1993; Laros, 2017; Lovett, 2023).

2. **Formative feedback**: I prioritise formative over purely summative assessment. This means integrating short feedback loops into teaching that help students monitor their progress and adjust their learning strategies (Aust et al., 2024; Black & Wiliam, 2009; Clark, 2012; Forsgren et al., 2018).

3. **Diversity and inclusion**: Teaching in international contexts has made me attentive to issues of gender, ethnicity, and cultural background. I strive to create learning environments where all perspectives are valued and respected, and where neurodiversity is an integral part, recognizing that students differ in how they process information, communicate, and demonstrate understanding (Rose, 2017; Schudde, 2018).

4. **Reducing cognitive barriers**: When working with more complex, math-based or statistical content, I place strong emphasis on reducing cognitive load and managing statistics anxiety. By offering a clear course structure, well-organised materials, and step-by-step scaffolding, I ensure that students' cognitive resources can be directed towards the actual learning tasks rather than being consumed by organisational or emotional barriers. This structured and supportive approach allows students to build confidence and maintain engagement, even when dealing with challenging quantitative content (Onwuegbuzie & Wilson, 2003; Sweller, 1994).

Teaching, for me, is not about transmitting knowledge in a one-directional way. It is about facilitating learning processes that are meaningful, sustainable, and connected to students' academic and professional trajectories. It is closely tied to students motivation and engagement, we as teachers need to recognise and facilitate (Ryan & Deci, 2000).

2. Teaching Experience

I have accumulated 723.5 hours of teaching experience across lectures, seminars, and thesis supervision. My teaching spans quantitative methods, computer-based data analysis, statistics, and research methodology.

- **2025** – Bachelor Thesis Preparation Course (SGEK03 Research Methods) [47h]
- **2025** – Introduction to Quantitative Approaches (SGEA23, Human Geography, 30 ECTS; UTVC24, Development Studies, 15 ECTS) [99h].
- **2023** – *Seminar: Quantitative Data Analysis* (Evaluation: 1.8).
- **2022** – *Seminar: Computer-based Data Analysis* (Evaluation: 1.3).
- **2022** – *Seminar: Quantitative Data Analysis* (Evaluation: 1.1).
- **2021** – *Seminar: Computer-based Data Analysis* (Evaluation: 1.6).
- **2021** – *Seminar: Quantitative Data Analysis* (Evaluation: 1.4).

In addition, I have supervised bachelor's theses in both education science and computer science, combining methodological guidance with academic writing support. Furthermore, I am engaging in informal PhD supervision.

Evaluations across courses and institutions are consistently strong, with most ratings between *1.1–1.6* (German system = *very good*).

3. Pedagogical Development

My teaching is continuously shaped by pedagogical training and reflection:

- **Teaching and Learning in Higher Education** (Lund University, 2025).
- **Teaching and Learning Research Methods in Social Science** (Lund University, 2025).
- **Didactics of Mathematics** (University of Tübingen, 2019).

These trainings deepened my understanding of constructive alignment, the role of formative assessment, and the challenges of teaching statistics and methods to students with heterogeneous backgrounds.

In practice, I have applied these insights by:

- Redesigning quantitative methods courses to focus on *progressive learning sequences* (breadth + depth) (Lovett, 2023).
- Integrating formative assessment strategies, including weekly feedback forms, to support learning autonomy (Clark, 2012).

- Experimenting with flipped classroom approaches (Baig & Yadegaridehkordi, 2023), where video inputs and self-study tasks free classroom time for application and discussion.

4. Innovation and Technology in Teaching

I see digital and AI-based tools not as replacements for teaching, but as enablers of new learning opportunities. Recent innovations in my courses include:

- **AI-assisted programming**: Using Google Colab, I introduced students to browser-based coding exercises. This approach removes technical barriers (installation, setup) and makes statistical programming in R or Python more accessible. Students can immediately focus on learning by doing (Bien & Mukherjee, 2025).
- **Responsible AI use**: I actively integrate discussions on AI tools such as ChatGPT into coursework, encouraging students to reflect on their opportunities and limitations in research practice (Kirkwood & Price, 2014).
- **Flipped learning**: I record short videos as input before seminars, allowing synchronous sessions to focus on discussion, clarification, and higher-order tasks (Baig & Yadegaridehkordi, 2023).
- **Adaptive learning environments**: I create modular structures in which students can select deeper exploration tasks depending on their interest and pace (Contrino et al., 2024).

By combining classical didactics with modern technologies, I help students build methodological skills while at the same time strengthening their critical reflection on the role of AI and digitalisation in the scientific process.

5. Teaching Style (Reflections from Student Feedback)

Student evaluations across multiple years and institutions consistently highlight key aspects of my teaching style:

- **Clarity and comprehensibility**: Students appreciate my calm and clear way of explaining complex methods. Many comment that statistical content, often perceived as abstract, becomes accessible and structured.
- **Supportive atmosphere**: The learning environment is described as open and respectful, with “no stupid questions.” This lowers anxiety around quantitative methods and encourages active participation.

- **Flexibility and responsiveness**: Students highlight my willingness to adapt sessions to their needs, revisit difficult content, and integrate feedback into subsequent classes.
- **Professional competence**: Evaluations consistently emphasise my expertise and preparedness. Students describe me as competent, approachable, and highly motivated to support learning.
- **Practical orientation**: Applied exercises in R and data analysis are seen as valuable for connecting theory to practice.
- **Critical engagement**: I encourage students to not only learn techniques but also question their assumptions, fostering a deeper methodological understanding.

Constructive feedback has also shaped my continuous improvement. Some students expressed a wish for even more scaffolded exercises, clearer alignment between lecture and seminar content, and shorter input phases with more interactive activities. I have taken these suggestions into account by expanding guided exercises, restructuring seminar flows, and increasing peer-to-peer interaction.

6. Future Teaching Vision

Looking ahead, I aim to further develop teaching that combines theoretical knowledge, quantitative training, and educational practice. In the context of digital education, I plan to address developments in artificial intelligence and immersive learning technologies. My goals include:

- Expanding the integration of **AI, data, and media literacy** into study curricula, preparing students for a future shaped by digital life and emerging technologies.
- Strengthening **interdisciplinary teaching**, where methods from data science are applied to disciplinary questions within education.
- Supporting **research-based teaching**, where students work with real data (e.g. internet, open datasets) to connect classroom learning with ongoing research.
- Building **inclusive teaching environments** that accommodate neurodiversity, diverse backgrounds and prior knowledge.
- Enhancing **mentorship and supervision** at BA, MA, and PhD levels, fostering independent research and critical engagement.

7. Conclusion

My teaching combines a reflective philosophy, strong didactic training, extensive experience in methods education, and continuous innovation with AI-based tools. The consistently

positive student evaluations confirm both the clarity and effectiveness of my teaching, as well as the supportive environment I aim to create.

In my new position, I would bring a profile that combines **excellent teaching evaluations, proven innovation, and pedagogical reflection**. My ambition is to inspire students to engage deeply with methods, embrace digital and AI tools critically and responsibly, and to foster a learning culture that is inclusive, exciting, and future-oriented.

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